National University of Computer & Emerging Sciences

Quiz #1(a) (Online)

Name & Roll Number:__________________ **Section:** ____
Date:__________________ **Maximum Marks 10**

1. A researcher records the heights (in cm) of 20 students in a university:

$$154, 160, 165, 170, 168, 175, 180, 155, 172, 178, 185, 167, 158, 162, 174, 169, 181, 177, 159, 171$$

Tasks:

(a) Use **Sturges' Rule** to determine the number of bins.

(b) Find the **class width** and construct class intervals.

(c) Determine **class boundaries**.

(d) Draw a **Histogram**, **Frequency Polygon**, and **Ogive**.

(e) Interpret the distribution (e.g., Is it skewed? Symmetric?).

National University of Computer & Emerging Sciences

Quiz # 1(b)

Name & Roll Number:____________________ **Section:** ____

Date:____________________ **Maximum Marks 10**

1. A household survey records the **monthly electricity bill** (in dollars) of 20 families:

$$45, 52, 67, 70, 58, 85, 90, 75, 80, 92, 100, 55, 60, 78, 66, 73, 88, 95, 62, 68$$

Tasks:

(a) Use **Sturges' Rule** to decide the number of bins.

(b) Find **class boundaries**.

(c) Construct a **Histogram**, **Frequency Polygon**, and **Ogive**.

(d) What is the **range** of the data?

(e) Comment on the **shape of the distribution**.

National University of Computer & Emerging Sciences

Quiz # 1(c)

Name & Roll Number:____________________ **Section:** ____
Date:____________________ **Maximum Marks 10**

1. A school records the **math exam scores** of 20 students:

$$40, 55, 72, 63, 79, 82, 95, 48, 67, 75, 88, 53, 69, 71, 84, 91, 58, 60, 77, 80$$

Tasks:

(a) Use **Sturges' Rule** to decide the number of bins.

(b) Find the **class width and boundaries**.

(c) Construct a **Histogram**, **Frequency Polygon**, and **Ogive**.

(d) How many students scored **above 80**?

(e) Is the data **normally distributed**?

National University of Computer & Emerging Sciences

Quiz # 1(d)

Name & Roll Number:____________________ **Section:** ____

Date:____________________ **Maximum Marks 10**

1. A fitness tracker records the **daily step count** of 20 people:

   $$3200, 4500, 5000, 7000, 5500, 6000, 5800, 6400, 7500, 8000, 8200, 3000, 4800, 5100, 7300, 7900, 6700$$

   Tasks:

   (a) Use **Sturges' Rule** to determine the number of bins.

   (b) Find **class boundaries**.

   (c) Draw a **Histogram**, **Frequency Polygon**, and **Ogive**.

   (d) What percentage of people walk **more than 7000 steps per day**?

   (e) Is the data **skewed**?